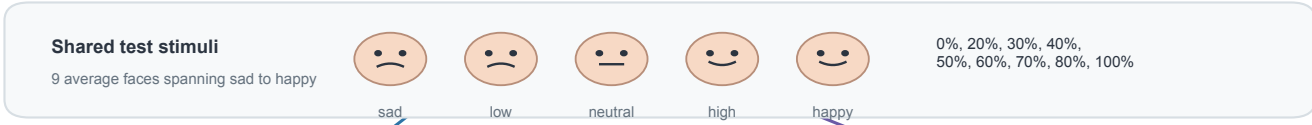
Overview of the experimental design and analysis workflow

The same morphed facial-expression stimuli link human behavioural data to CNN evaluation.



Human behavioural benchmark

- 40 participants
  normal/corrected vision
- 2AFC task
  F: happy J: sad
- Random order
  9 faces × 20
  = 180 trials
- Psychometric function → PSE and response probability

CNN modelling pipeline

- Pre-training
  ImageNet or
  VGGFace2
- Transfer learning
  AffectNet
  emotion task
- Same 9 faces
  Softmax output
  probability
- Psychometric function → PSE, RMSE and fit quality

Human–model comparison: behavioural similarity

Experiment 1

Compression strength

Late-stage SE insertion
R = 2, 4, 8, 16, 32
vs. AlexNet and VGG16

Experiment 2

Insertion position

Early, middle, or late
SE insertion × pre-training
× compression ratio

Experiment 3

Feature-use analysis

Grad-CAM and ROI saliency
Full face vs. eye, nose,
and mouth occlusion

Compression strength and insertion stage determine behavioural similarity and model feature use.